ECON 4720: HA03

Rahul Sunilkumar

2025-10-24

Problem 1

Are the following vectors linearly independent?

1a

$$a_1 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 7 \\ 6 \end{pmatrix}, \quad a_3 = \begin{pmatrix} 12 \\ 10 \end{pmatrix}$$

Solution Attempt These vectors live in $\mathbb{R}^2$, so any set of three vectors must be linearly dependent. To make this explicit, solve

$$c_1a_1 + c_2a_2 + c_3a_3 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \iff \quad \begin{cases} 3c_1 + 7c_2 + 12c_3 = 0, \\ 2c_1 + 6c_2 + 10c_3 = 0. \end{cases}$$

Row-reducing (work condensed) gives

$$\begin{bmatrix} 3 & 7 & 12 \\ 2 & 6 & 10 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{3}{2} \end{bmatrix},$$

so

$$c_1 = -\tfrac{1}{2}\, c_3, \qquad c_2 = -\tfrac{3}{2}\, c_3,$$

with c_3 free. Taking $c_3 = 1$ yields the concrete dependence

$$-\tfrac{1}{2}\, a_1 - \tfrac{3}{2}\, a_2 + 1 \cdot a_3 = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

and, clearing denominators,

$$-1 \cdot a_1 - 3 \cdot a_2 + 2 \cdot a_3 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

$$\boxed{\text{Therefore, the vectors are linearly dependent (not independent).}}$$

$\square$

1b

$$a_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad a_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Solution Attempt Test

$$c_1 a_1 + c_2 a_2 + c_3 a_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Writing components,

$$c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Thus

$$c_1 = 0, \quad c_2 = 0, \quad c_3 = 0,$$

so the only solution is the trivial one.

$$\boxed{\text{Therefore, the vectors are linearly independent.}}$$

$\square$

Problem 2

Find determinants of the following matrices:

$$A = \begin{bmatrix} 1 & 5 & 7 \\ 3 & 2 & 8 \\ 6 & 1 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 3 & 2 \\ 6 & -2 & 3 \\ 7 & 10 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 13 & 4 & 1 \\ 0 & 4 & 1 \\ -7 & 2 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} -4 & 4 & 12 \\ 3 & -3 & -9 \\ 8 & 2 & 6 \end{bmatrix}$$

Solution Attempt Cofactor along row 1:

$$\det A = 1 \begin{vmatrix} 2 & 8 \\ 1 & 9 \end{vmatrix} - 5 \begin{vmatrix} 3 & 8 \\ 6 & 9 \end{vmatrix} + 7 \begin{vmatrix} 3 & 2 \\ 6 & 1 \end{vmatrix} = 1(18-8) - 5(27-48) + 7(3-12) = 10 + 105 - 63 = 52.$$

Cofactor along row 1:

$$\det B = -1\begin{vmatrix}-2 & 3\\ 10 & 0\end{vmatrix} - 3\begin{vmatrix}6 & 3\\ 7 & 0\end{vmatrix} + 2\begin{vmatrix}6 & -2\\ 7 & 10\end{vmatrix} = -1(-30)-3(-21)+2(60+14) = 30+63+148 = 241.$$

Expand down column 1 (uses the zero):

$$\det C = 13\begin{vmatrix}4 & 1\\ 2 & 3\end{vmatrix} + 0\cdot(\cdots) + (-7)\begin{vmatrix}4 & 1\\ 4 & 1\end{vmatrix} = 13(12-2) + (-7)(0) = 130.$$

Note $R_2 = (-\frac{3}{4})R_1$ (rows dependent), so

$$\det D = 0.$$

$$\boxed{\det A = 52, \quad \det B = 241, \quad \det C = 130, \quad \det D = 0.}$$

□

Problem 3

Using definitions, check whether the following matrices are positive definite or positive semidefinite:

Problem 3a

$$A = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}, \quad B = \begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}, \quad C = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}, D = \begin{bmatrix}1 & -3\\ 0 & 1\end{bmatrix}, \quad E = \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}$$

Solution Attempt Using the definition: a matrix M is PSD if $x^\top Mx \geq 0$ for all $x \neq 0$, and PD if $x^\top Mx > 0$ for all $x \neq 0$.

$A = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

$$x^\top Ax = x_1^2 + x_2^2 > 0 \text{ for all } x \neq 0.$$

$\Rightarrow$ **PD**.

$B = \begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}$

$$x^\top Bx = x_1^2 \geq 0, \quad \text{and } x^\top Bx = 0 \text{ for } x = (0,1)^\top \neq 0.$$

$\Rightarrow$ **PSD, not PD**.

$C = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$$x^\top C x = 2x_1x_2.$$

Takes both signs (e.g., $x = (1,-1)^\top \Rightarrow -2$ and $x = (1,1)^\top \Rightarrow 2$).
$\Rightarrow$ **indefinite** (neither PSD nor PD).

$D = \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}$

$$x^\top D x = x_1^2 - 3x_1x_2 + x_2^2 = \left(x_1 - \tfrac{3}{2}x_2\right)^2 - \tfrac{5}{4}x_2^2.$$

Negative for $x = (1,2)^\top$ $(1 - 6 + 4 = -1)$, positive for $x = (1,0)^\top$ $(= 1)$.
$\Rightarrow$ **indefinite**.

$E = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

$$x^\top E x = x_1x_2,$$

which takes both signs (e.g., $x = (1,-1)^\top \Rightarrow -1$, $x = (1,1)^\top \Rightarrow 1$).
$\Rightarrow$ **indefinite**.

$$\boxed{A: \text{ PD}; \quad B: \text{ PSD (not PD)}; \quad C, D, E: \text{ indefinite.}}$$

$\square$

Problem 3b

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -4 & 2 \\ -1 & 2 & -1 \end{bmatrix}$$

Solution Attempt Using the definition: M is PSD if $x^\top M x \geq 0$ for all $x \neq 0$, and PD if $x^\top M x > 0$ for all $x \neq 0$.

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \text{ (symmetric)}$$

$$x^\top A x = 2x_1x_3 + x_2^2.$$

Takes both signs (e.g., $x = (1, 0, -1)^\top \Rightarrow -2$, $x = (1, 0, 1)^\top \Rightarrow 2$).

$$\Rightarrow \text{indefinite (neither PSD nor PD).}$$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 0 \end{bmatrix} \text{ (symmetric)}$$

Test by quadratic form:

$$x = (1, -2, 1)^\top \;\Rightarrow\; x^\top Bx = -9 < 0, \qquad x = (1, 1, 1)^\top \;\Rightarrow\; x^\top Bx = 27 > 0.$$

So it attains both signs.

$$\Rightarrow \text{indefinite (neither PSD nor PD).}$$

$$C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \text{ (symmetric)}$$

$$x^\top Cx = x_1^2 + x_2^2 + x_3^2 + 2x_1x_3 = (x_1 + x_3)^2 + x_2^2 \geq 0,$$

and equals 0 for $x = (1, 0, -1)^\top \neq 0$.

$$\Rightarrow \text{PSD (not PD).}$$

$$D = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -4 & 2 \\ -1 & 2 & -1 \end{bmatrix} \text{ (symmetric)}$$

Observe

$$D = -\,vv^\top, \quad v = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}.$$

Hence

$$x^\top Dx = -\,(v^\top x)^2 \leq 0, \text{ with } x = v \Rightarrow x^\top Dx < 0.$$

So it is negative semidefinite (nonpositive for all x).

$$\Rightarrow \text{not PSD/PD in the positive sense (NSD).}$$

$$\boxed{A: \text{ indefinite}, \quad B: \text{ indefinite}, \quad C: \text{ PSD (not PD)}, \quad D: \text{ negative semidefinite (not PSD/PD).}}$$

$\square$

Problem 4

Find the rank of each the following matrices:

$$A = \begin{bmatrix} 3 & 6 & 4 & 8 \\ 2 & 7 & 1 & 9 \\ 4 & 2 & 5 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 18 & 2 \\ 7 & -4 \\ 6 & 11 \end{bmatrix}, \quad C = \begin{bmatrix} -5 & 8 & 11 & 0 \\ 13 & 3 & 0 & 2 \\ 10 & 0 & -6 & 2 \end{bmatrix}$$

Solution Attempt I'll show extended work for this one and clean it up for the rest of the parts.

$$\begin{bmatrix} 3 & 6 & 4 & 8 \\ 2 & 7 & 1 & 9 \\ 4 & 2 & 5 & 0 \end{bmatrix} \xrightarrow{R_2 \leftarrow 3R_2 - 2R_1,\ R_3 \leftarrow 3R_3 - 4R_1} \begin{bmatrix} 3 & 6 & 4 & 8 \\ 0 & 9 & -5 & 11 \\ 0 & -18 & -1 & -32 \end{bmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 + 2R_2} \begin{bmatrix} 3 & 6 & 4 & 8 \\ 0 & 9 & -5 & 11 \\ 0 & 0 & -11 & -10 \end{bmatrix} \xrightarrow{R_3 \leftarrow -\frac{1}{11}R_3,\ R_2 \leftarrow \frac{1}{9}R_2} \begin{bmatrix} 3 & 6 & 4 & 8 \\ 0 & 1 & -\frac{5}{9} & \frac{11}{9} \\ 0 & 0 & 1 & \frac{10}{11} \end{bmatrix}$$

$$\xrightarrow{R_2 \leftarrow R_2 + \frac{5}{9}R_3,\ R_1 \leftarrow R_1 - 4R_3} \begin{bmatrix} 3 & 6 & 0 & \frac{48}{11} \\ 0 & 1 & 0 & \frac{19}{11} \\ 0 & 0 & 1 & \frac{10}{11} \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1 - 6R_2,\ R_1 \leftarrow \frac{1}{3}R_1} \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & \frac{19}{11} \\ 0 & 0 & 1 & \frac{10}{11} \end{bmatrix}.$$

There are 3 pivots, so

$$\mathrm{rank}(A) = 3.$$

(Equivalently, the 3×3 left block has determinant $-33 \neq 0$.)

$$B = \begin{bmatrix} 18 & 2 \\ 7 & -4 \\ 6 & 11 \end{bmatrix}$$

Condensed row-reduction:

$$\begin{bmatrix} 18 & 2 \\ 7 & -4 \\ 6 & 11 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Two pivots $\Rightarrow \mathrm{rank}(B) = 2$.

(Equivalently, the 2×2 minor $\begin{vmatrix} 18 & 2 \\ 7 & -4 \end{vmatrix} = -86 \neq 0$.)

$$C = \begin{bmatrix} -5 & 8 & 11 & 0 \\ 13 & 3 & 0 & 2 \\ 10 & 0 & -6 & 2 \end{bmatrix}$$

Condensed row-reduction:

$$\begin{bmatrix} -5 & 8 & 11 & 0 \\ 13 & 3 & 0 & 2 \\ 10 & 0 & -6 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & \frac{5}{64} \\ 0 & 1 & 0 & \frac{21}{64} \\ 0 & 0 & 1 & -\frac{13}{64} \end{bmatrix}.$$

Three pivots $\Rightarrow \operatorname{rank}(C) = 3$.

(Equivalently, $\det \begin{bmatrix} -5 & 8 & 11 \\ 13 & 3 & 0 \\ 10 & 0 & -6 \end{bmatrix} = 384 \neq 0$.)

$$\boxed{\operatorname{rank}(A) = 3, \quad \operatorname{rank}(B) = 2, \quad \operatorname{rank}(C) = 3.}$$

□

Problem 5

Find such value of the parameter α that the following vectors are linearly dependent:

$$a_1 = \begin{pmatrix} 2 \\ \alpha \\ -2 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}, \quad a_3 = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$$

Solution Attempt The vectors are linearly dependent if the determinant of the matrix formed by placing them as columns equals 0:

$$\det \begin{bmatrix} 2 & 3 & -1 \\ \alpha & 0 & 2 \\ -2 & 4 & 1 \end{bmatrix} = 0.$$

Compute the determinant by expansion along the first row:

$$2 \begin{vmatrix} 0 & 2 \\ 4 & 1 \end{vmatrix} - 3 \begin{vmatrix} \alpha & 2 \\ -2 & 1 \end{vmatrix} + (-1) \begin{vmatrix} \alpha & 0 \\ -2 & 4 \end{vmatrix} = 0.$$

Simplify each minor:

$$2(0 \cdot 1 - 8) - 3(\alpha \cdot 1 - (-4)) - 1(\alpha \cdot 4 - 0) = 0,$$

$$2(-8) - 3(\alpha + 4) - (4\alpha) = 0,$$

$$-16 - 3\alpha - 12 - 4\alpha = 0,$$
$$-7\alpha - 28 = 0.$$

Hence,

$$\alpha = -4.$$

$$\boxed{\alpha = -4}$$

□

Problem 6

Solve the following systems of linear questions:

Problem 6a

$$\begin{cases} x + y + z = 11 \\ x - y - z = -3 \\ -x + y - z = 5 \\ 3x - y + 2z = 2 \end{cases}$$

Solution Attempt Add the first two equations to eliminate y, z:

$$(x + y + z) + (x - y - z) = 11 + (-3) \;\Rightarrow\; 2x = 8 \;\Rightarrow\; x = 4.$$

Use $x = 4$ in the 2nd and 3rd equations:

$$4 - y - z = -3 \;\Rightarrow\; y + z = 7, \qquad -4 + y - z = 5 \;\Rightarrow\; y - z = 9.$$

Solve the 2×2 system:

$$y = \tfrac{(7+9)}{2} = 8, \qquad z = 7 - y = -1.$$

Check in the 4th equation: $3(4) - 8 + 2(-1) = 12 - 8 - 2 = 2.$

$$\boxed{(x, y, z) = (4,\ 8,\ -1).}$$

□

Problem 6b

$$\begin{cases} x+y+z+t+p=17 \\ x-y-z-t-p=-5 \\ z+t+p+y=11 \\ p-x-y=1 \\ -t+x=10 \end{cases}$$

Solution Attempt From the last two equations,

$$t = x - 10, \qquad p = 1 + x + y.$$

Then $y + z + t + p = 11 \Rightarrow 2x + 2y + z = 20 \Rightarrow z = 20 - 2x - 2y$.
Plugging into $x + y + z + t + p = 17$ gives $x = 6$, hence

$$t = -4, \qquad p = 7 + y, \qquad z = 8 - 2y.$$

Explicit check of the 2nd equation:

$$x - y - z - t - p = 6 - y - (8 - 2y) - (-4) - (7 + y) = -5.$$

Thus the solution family is

$$\boxed{(x, y, z, t, p) = \big(6,\ y,\ 8 - 2y,\ -4,\ 7 + y\big), \quad y \in \mathbb{R}.}$$

$\square$

Problem 6c

$$\begin{cases} x+y+z+t=-6 \\ x-y-z-t=20 \\ y-x=-39 \\ 2x+3t+y-4z=32 \end{cases}$$

Solution Attempt From $y - x = -39$ we get

$$y = x - 39.$$

Substitute into the first two equations:

$$x + (x - 39) + z + t = -6 \ \Rightarrow\ 2x + z + t = 33, x - (x - 39) - z - t = 20 \ \Rightarrow\ z + t = 19.$$

Combine to solve for x:

$$2x + (z + t) = 33 \ \Rightarrow\ 2x + 19 = 33 \ \Rightarrow\ x = 7, \quad y = x - 39 = -32.$$

Use $z + t = 19$ and the last equation:

$$2x + 3t + y - 4z = 32 \ \Rightarrow\ 14 + 3t - 32 - 4z = 32 \ \Rightarrow\ 3t - 4z = 50.$$

With $t = 19 - z$:

$$3(19 - z) - 4z = 50 \ \Rightarrow\ 57 - 7z = 50 \ \Rightarrow\ z = 1, \quad t = 18.$$

$$\boxed{(x, y, z, t) = (7,\ -32,\ 1,\ 18).}$$

□

Problem 6d

$$\begin{cases} -4x - 5y - 5z = -74 \\ -3x - 5y - z = -4 \\ -x + 7y + 6z = 48 \end{cases}$$

From the first two equations, eliminate y by subtraction:

$$(-4x - 5y - 5z) - (-3x - 5y - z) = -74 - (-4) \ \Rightarrow\ -x - 4z = -70 \ \Rightarrow\ x = 70 - 4z.$$

Use the third equation to relate y and z:

$$-(70 - 4z) + 7y + 6z = 48 \ \Rightarrow\ 7y + 10z = 118 \ \Rightarrow\ y = \frac{118 - 10z}{7}.$$

Substitute x and y into $-3x - 5y - z = -4$:

$$-3(70 - 4z) - 5\left(\frac{118 - 10z}{7}\right) - z = -4.$$

Multiply by 7 to clear denominators and solve for z (work condensed):

$$-1302 + 129z = -28 \ \Rightarrow\ z = 16.$$

Then

$$x = 70 - 4 \cdot 16 = 6, \qquad y = \frac{118 - 10 \cdot 16}{7} = -6.$$

$$\boxed{(x, y, z) = (6,\ -6,\ 16).}$$

□

Problem 6e

$$\begin{cases} x + 2y - z + 3t = 0 \\ x + 2y + 2z - t = 1 \\ 3x + 6y + 5t = 1 \\ 4x + 8y - z + 8t = 1 \end{cases}$$

Solution Attempt Subtract the first equation from the second:

$$(x + 2y + 2z - t) - (x + 2y - z + 3t) = 1 - 0 \Rightarrow 3z - 4t = 1 \Rightarrow z = \frac{1 + 4t}{3}.$$

Use this in the first equation to get x:

$$x + 2y - \frac{1 + 4t}{3} + 3t = 0 \Rightarrow x = -2y + \frac{1 - 5t}{3}.$$

With these x, z, the third and fourth equations are automatically satisfied (they reduce to identities), so y and t are free parameters.

Therefore the solution set is

$$(x, y, z, t) = \Big(\frac{1 - 5t}{3} - 2y,\ y,\ \frac{1 + 4t}{3},\ t\Big), \quad y, t \in \mathbb{R}.$$

Equivalently, in vector form,

$$\begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} \frac{1}{3} \\ 0 \\ \frac{1}{3} \\ 0 \end{pmatrix} + y \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -\frac{5}{3} \\ 0 \\ \frac{4}{3} \\ 1 \end{pmatrix}.$$

$$\boxed{\text{Infinitely many solutions with } y, t \text{ free.}}$$

□

Problem 7

For which values of λ the following matrix is invertible?

$$A = \begin{bmatrix} 1 & -2 & 2 \\ \lambda & 3 & 0 \\ 2 & 1 & 1 \end{bmatrix}$$

Solution Attempt A is invertible iff $\det A \neq 0$. Compute

$$\det A = 1 \begin{vmatrix} 3 & 0 \\ 1 & 1 \end{vmatrix} - (-2) \begin{vmatrix} \lambda & 0 \\ 2 & 1 \end{vmatrix} + 2 \begin{vmatrix} \lambda & 3 \\ 2 & 1 \end{vmatrix} = 3 + 2\lambda + 2(\lambda - 6) = 4\lambda - 9.$$

$$\boxed{A \text{ is invertible iff } 4\lambda - 9 \neq 0 \iff \lambda \neq \tfrac{9}{4}.}$$

$\square$

Problem 8

Find all values a, for which the system of linear equations does not have any solution.

$$\begin{cases} 2x + (9a^2 - 2)y = 3a \\ x + y = 1 \end{cases}$$

Solution Attempt The coefficient matrix is

$$M = \begin{bmatrix} 2 & 9a^2 - 2 \\ 1 & 1 \end{bmatrix}, \qquad \det M = 2 \cdot 1 - (9a^2 - 2) \cdot 1 = 4 - 9a^2.$$

If $\det M \neq 0$ (i.e., $a \neq \pm\frac{2}{3}$), there is a unique solution. If $\det M = 0$ (i.e., $a = \pm\frac{2}{3}$), check consistency:

For $a = \frac{2}{3}$:

$$2x + (4-2)y = 2 \iff 2x + 2y = 2, \quad \text{and} \quad x + y = 1.$$

These are the same equation $\Rightarrow$ infinitely many solutions (consistent).

For $a = -\frac{2}{3}$:

$$2x + 2y = -2, \quad \text{but} \quad x + y = 1 \Rightarrow 2x + 2y = 2,$$

which is a contradiction $\Rightarrow$ no solution.

$$\boxed{\text{No solution exactly when } a = -\tfrac{2}{3}.}$$

$\square$

Problem 9

Find a value of b such that system of linear questions has infinitely many solutions:

$$\begin{cases} 6x + 2by = 3 \\ 4x + 2y = 2 \end{cases}$$

Solution Attempt A system of two linear equations in two variables has infinitely many solutions when one equation is a scalar multiple of the other.

That is, we must have proportional coefficients:

$$\frac{6}{4} = \frac{2b}{2} = \frac{3}{2}.$$

Simplify:

$$\frac{6}{4} = \frac{3}{2}, \quad \frac{2b}{2} = b, \quad \frac{3}{2} = \frac{3}{2}.$$

So we require $b = \frac{3}{2}$.

$$\boxed{b = \tfrac{3}{2}.}$$

$\square$

Problem 10

For the following matrices check the statement: "If a matrix X has full rank then the matrix $X'X$ is invertible" and find the inverse $(X'X)^{-1}$:

Problem 10a

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Solution Attempt If X has full column rank, then $X'X$ is invertible.

Here $X \in \mathbb{R}^{4\times 2}$ and $\mathrm{rank}(X) = 2$ (its two columns are the standard basis vectors in $\mathbb{R}^4$), so X has full column rank. Compute

$$X'X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2,$$

which is invertible.

$$\boxed{(X'X)^{-1} = I_2.}$$

□

Problem 10b

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}$$

Solution Attempt Columns:

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \qquad x_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}.$$

They are linearly independent, so $\mathrm{rank}(X) = 2$ (full column rank) and thus $X'X$ is invertible.

Compute

$$X'X = \begin{bmatrix} x_1^\top x_1 & x_1^\top x_2 \\ x_2^\top x_1 & x_2^\top x_2 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \det(X'X) = 4 \cdot 1 - 1 \cdot 1 = 3 \neq 0.$$

Hence

$$(X'X)^{-1} = \frac{1}{3} \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix}.$$

$$\boxed{X'X = \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix}, \quad (X'X)^{-1} = \tfrac{1}{3} \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix}.}$$

□

Problem 10c

$$X = \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 0 \\ 4 & 1 \end{bmatrix}$$

Solution Attempt Columns are independent (clearly not multiples), so $\mathrm{rank}(X) = 2$ (full column rank) and $X'X$ is invertible. Compute

$$X'X = \begin{bmatrix} x_1^\top x_1 & x_1^\top x_2 \\ x_2^\top x_1 & x_2^\top x_2 \end{bmatrix} = \begin{bmatrix} 1^2 + 2^2 + 3^2 + 4^2 & 1 \cdot 1 + 2 \cdot 3 + 3 \cdot 0 + 4 \cdot 1 \\ \cdot & 1^2 + 3^2 + 0^2 + 1^2 \end{bmatrix} = \begin{bmatrix} 30 & 11 \\ 11 & 11 \end{bmatrix}.$$

Its determinant is $\det(X'X) = 30 \cdot 11 - 11 \cdot 11 = 209 \neq 0$, so

$$(X'X)^{-1} = \frac{1}{209} \begin{bmatrix} 11 & -11 \\ -11 & 30 \end{bmatrix}.$$

$$\boxed{X'X = \begin{bmatrix} 30 & 11 \\ 11 & 11 \end{bmatrix}, \quad (X'X)^{-1} = \tfrac{1}{209} \begin{bmatrix} 11 & -11 \\ -11 & 30 \end{bmatrix}.}$$

□